

Victoria You

(615) 484-1850 | vyou2@illinois.edu

victoriayou3306.github.io | <https://www.linkedin.com/in/victoria-you/>

EDUCATION

University of Illinois Urbana-Champaign

Expected May 2028

3.93/4.00; B.S Mechanical Engineering; B.S Computational Physics; Minor in Semiconductor Engineering

- Dean's List; James Scholar Honors; Pi Tau Sigma; DaRin Butz Research Scholar; Engineering Visionary Scholarship; Samsung Semiconductor Technology Scholarship
- Relevant Coursework: Semiconductor Devices; Atomic Physics & Quantum Theory; Robotic Manipulators; Mechanical Design; Fluid Dynamics; Classical Mechanics; Thermodynamics; Solid Mechanics; Electrical and Electronic Circuits

RESEARCH & EXPERIENCE

Hooverman High Energy Group

Jul 2026 – *Present*

Undergraduate Research

Champaign, IL

- Facilitating product development of muon tomography tools from lab-scale iterative prototyping to field deployment
- Designing hermetically sealed, corrosion-resistant enclosures for oil borehole installation of depths up to 1 km

KLA

May 2026 - *Present*

Mechatronics Intern

Ann Arbor, MI

- Developed a Python based system identification pipeline for generating precision motion stage models for controls
- Automated the evaluation of different excitation and processing methods to build Frequency Response Functions
- Validated an array of 700+ models, comparing simulated vs experimental trajectories in a range of 1 μm – 5 cm
- Designed a live update GUI for loop shaping to visualize system stability margins using plant & controller Bode Plots

UIUC Mechanical Science & Engineering

Jan 2026- Jun 2026

Web Content Developer

Champaign, IL

- Developing and maintaining online reference pages for Design for Manufacturability coursework and materials
- Typesetting lecture slides as technical documents in LaTeX and generating supplemental examples and figures
- Managing version control and collaborative workflows with GitHub while optimizing functionality and accessibility

MIT Lincoln Laboratory

Sep 2025- Jan 2026

Microfabrication Process Engineering Co-Op

Lexington, MA

- Increased multi-wafer uniformity of HZO Atomic Layer Deposition films by 88% with iterative recipe optimization
- Identified an 8% ellipsometry measurement error via XRR cross-validation and generated revised optical models
- Refined stability and conditioning for metals in E-beam Evaporation, improving Hf deposition conductivity by 49%
- Developed Selective Si Epitaxial Growth processes for in-house fabrication of RSD transistors and sub-90nm CMOS

Cygnus Photonics, Inc.

Feb 2025-Aug 2025

Research & Development Engineering Intern

Champaign, IL

- Manufactured UV/VUV exposure systems for customer research and performed post-use troubleshooting and repair
- Completed incoming quality control to drive design revisions; created 25+ technical part drawings for manufacturing
- Designed and iterated modular experimentation units from 3D-printed and laser cut prototypes to machined products
- Reprogrammed a 3D-Printing Gantry using LabVIEW to determine uniformity for lamp area increases of up to 500%

Nashville Motorcycle Repair

May 2024-Aug 2024

Restoration Mechanic Intern

Nashville, TN

- Fully restored non-running motorcycles for resale: Honda Z50 Minibike; 1982 Suzuki GS1100EZ
- Partially restored motorcycles in poor condition for resale: 1996 Moto Guzzi Sport 1000; Suzuki LS650 Savage Rycar
- Rebuilt and serviced all customer Ural Motorcycle wheels affected by the 2011-2017 Ural Factory Rim Recalls

LEADERSHIP & INVOLVEMENT

Product Development Committee Co-director, American Society of Mechanical Engineers @Illinois Apr 2026-Present

- Facilitating 4 concurrent engineering projects spanning multi-semester design cycles for industry & academic clients
- Running design review cycles with undergraduate sub-teams spanning mechanical, thermal, and controls domains
- Authored technical onboarding materials and preliminary design resources tailored to each project

NovoPrint Multi-Extruder Arm Project Lead, American Society of Mechanical Engineers @Illinois Jan 2025-Present

- Leading a team of 15+ undergraduates to design and manufacture two 6-Axis robot arms for non-planar 3D-printing
- Path planning and simulating motion using ROS Rviz, controlling via CAN bus for simultaneous multi-arm printing
- Awarded 3rd Place Distinguished Technology Award at the 2025 Illinois Engineering Open House

SKILLS & INTERESTS

- Software: MATLAB, JMP, Java, Python, Autodesk Inventor, ROS, C++, Arduino, SolidWorks, Autodesk Fusion 360
- Manufacturing: Rapid Prototyping, FDM & SLA 3D Printing, Laser Cutting, Soldering, TIG Welding
- Nanofabrication: Thin-Film Deposition (ALD, E-beam Evaporation), Si Epitaxy, Ellipsometry, Profilometry, SEM
- Hobbies & Interests: Varsity Illini Rowing, Motorcycles, Snowboarding, Origami Paper Cranes